JPE Replication Report

PUBLISHED
April 27, 2026

1 Paper Identification

ID	20250266
Author	Sprenger
Title	Connecting Common Ratio and Common Consequence Preferences
Iteration	2

Some useful information up front:

- JPE Data and Code Availability Policy
- Step by step guidance from Data Editor for Authors
- Template README

2 Summary

In assessing compliance with our Data and Code Availability Policy, our reproducibility team has identified the following issues, which we ask you to address:

3 General

- [x] Data Availability and Provenance Statements
  - [x] Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
  - [x] Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
  - [x] License for Data (optional, but recommended)
  - [x] Details on each Data Source
- [x] Dataset list
- [x] Computational requirements
  - [x] Software Requirements
  - [x] Controlled Randomness (as necessary)
  - [x] Memory, Runtime, and Storage Requirements
- [x] Description of programs/code

- [x] License for Code (Optional, but recommended)
- [x] List of tables and programs
- [x] References (Optional, but recommended)

4 High-level Description of Research Project

This research project…

- [ ] uses observational data.
- [ ] uses *confidential* observational data.
- [x] uses data generated via experiment or trial by the authors (in field or lab).
  - [ ] For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
  - [x] A PDF copy of IRB approval is contained in the package.
  - [ ] A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
    - [x] Details outlining the design of the experiment, including information on the selection and eligibility of participants is included in the README.
  - [x] A PDF copy of the instructions given to participants in both original language and an English translation is included.
- [x] uses data generated via survey by the authors (in field or online)
  - [x] The programs used to administer the survey (e.g. qualtrics survey file) are included.
  - [x] A PDF copy of IRB approval is contained in the package.
  - [x] A PDF of the original questionaire(s) and information on the selection and eligibility of participants is included.
- [ ] is a simulation study: the data are generated by an algorithm.
- [ ] is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
raw-data/qualtrics-raw.csv	yes	no	no	no	yes	no
raw-data/demographics-anonymized.csv	yes	no	no	no	yes	no
raw-data/Published-Blat-CCE-data-2022-	yes	no	no	yes	yes	yes

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
AEJMicro.csv						
raw-data/Published-Blat-CRE-data-2023-EE.csv	yes	no	no	yes	yes	yes
raw-data/[dataset]_definitions.csv	yes	no	no	no	no	no

5.1.1 Qualtrics Data

- [x] Dataset is provided in public deposit
- [ ] Dataset is PRIVATELY provided (not for publication)
- [ ] DOI or URL is provided, and works
- [ ] Access conditions are described:
  - Primary data collected by the authors
- [ ] Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- [x] Data are cited
  - [ ] in the manuscript
  - [x] in the README:

> (Primary data) Author-collected online experiment programmed in Qualtrics and administered to participants recruited on Prolific (October 5, 2022; see experimental documentation in experimental-materials/ and the pre-registration listed above). Raw data: raw-data/qualtrics-raw.csv and raw-data/demographics-anonymized.csv.

5.1.2 Anonymized Demographics

- [x] Dataset is provided in public deposit
- [ ] Dataset is PRIVATELY provided (not for publication)
- [ ] DOI or URL is provided, and works
- [ ] Access conditions are described:
  - Primary data collected by the authors
- [ ] Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- [x] Data are cited
  - [ ] in the manuscript
  - [x] in the README:

> (Primary data) Author-collected online experiment programmed in Qualtrics and administered to participants recruited on Prolific (October 5, 2022; see experimental documentation in experimental-materials/ and the pre-registration listed above). Raw data: raw-data/qualtrics-raw.csv and raw-data/demographics-anonymized.csv.

5.1.3 Dataset Definitions

- [x] Dataset is provided in public deposit
- [ ] Dataset is PRIVATELY provided (not for publication)
- [ ] DOI or URL is provided, and works
- [ ] Access conditions are described:
  - Author generated defintions for variables
- [ ] Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- [ ] Data are cited
  - [ ] in the manuscript
  - [ ] in the README

5.1.4 Published-Blat-CCE-data-2022-AEJMicro.csv

- [x] Dataset is provided in public deposit
- [ ] Dataset is PRIVATELY provided (not for publication)
- [ ] DOI or URL is provided, and works
- [x] Access conditions are described:
  - Created from tables included in previously published journal papers.
  - Included in the data section to match the README
- [x] Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- [x] Data are cited
  - [x] in the manuscript
  - [x] in the README

> Blavatskyy, P., Ortmann, A., & Panchenko, V. (2022). On the experimental robustness of the Allais paradox. American Economic Journal: Microeconomics, 14(1), 143–163.

5.1.5 Published-Blat-CRE-data-2022-EE.csv

- [x] Dataset is provided in public deposit
- [ ] Dataset is PRIVATELY provided (not for publication)
- [ ] DOI or URL is provided, and works
- [x] Access conditions are described:
  - Created from tables included in previously published journal papers. Authors made minor corrections that are described in the README.
  - Included in the data section to match the README
- [x] Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- [x] Data are cited
  - [x] in the manuscript
  - [x] in the README

> Blavatskyy, P., Panchenko, V., & Ortmann, A. (2023). How common is the common-ratio effect? Experimental Economics, 26(2), 253–272.

6 Requirements

- [x] README is in TXT, MD, or PDF format
- [x] README is accessible at the root of the package (not inside a folder)
- [x] Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- [x] Authors (with affiliations) are listed in the same order as on the paper

6.1 Stated Requirements

- [ ] No requirements specified
- [x] Software Requirements specified as follows:
  - Stata MP 19.5
    - Stata Packages: estout, unique, binscatter
  - Matlab: 25.1.0.2943329 (R2025a)
  - R version 4.5.1
    - R packages: stats4, mvtnorm, dplyr, kableExtra
- [x] Computational Requirements specified as follows:
  - <25MB of storage space required
- [x] Time Requirements specified as follows:
  - <10 minutes for Stata/Matlab-only
  - 10-120 minutes for full replication including R
- [x] Requirements are complete.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals*. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 3 - Variables flagged in data: 5 - Code files with PII references: 17 - PII references in code: 391

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	`README.txt`	1	lat

File Type	File	Variables/References	PII Categories
Data	`demographics-anonymized.csv`	2	sex, country
Data	`qualtrics-raw.csv`	2	son, block, loc
Code	`clean_data.do`	55	block, loc, son, lat, lon, name, second, school, gender, sex, country
Code	`functions.R`	11	lat
Code	`mainR.R`	7	lat, son, name, minute, second
Code	`mainS.do`	5	second, loc, name, lon
Code	`makeAppendixFigures.do`	20	loc, lat
Code	`makeAppendixTables.do`	102	lat, sex, degree, country, loc, lname, name, minute, lon
Code	`makeFigure1.m`	17	son, lat, loc, location
Code	`makeFigure9.m`	10	son, name, lat
Code	`makeFigureD1.R`	1	lat
Code	`makeFigureE1.m`	10	son, lat
Code	`makeFigures3-7.do`	43	loc, lat, lon, name
Code	`makeTable2.do`	9	lat, loc, name
Code	`makeTableD1.R`	3	lat, name
Code	`makeTableD2.R`	3	lat, name
Code	`reducedForm.R`	27	lon, lat, lname, name
Code	`statsInText.do`	13	son, lon, loc, lat
Code	`structuralEstimates.m`	55	son, name, lat, block, loc, location, degree

See [Appendix] for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-04-15T19:47:02.344

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (`C:\` etc)
28	1	10	2	0

7.1.3 Duplicate Files Report

We did not find any duplicate files.

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We did not find any files larger than 100MB.

7.2 Code description

cloc	github.com/AlDanial/cloc v 2.02 T=0.06 s (610.3 files/s, 154875.3 lines/s)

Language	files	blank	comment	code
CSV	8	0	0	4502
Stata	7	451	143	1531
MATLAB	15	266	250	1120
R	6	151	164	616
Text	2	112	0	337

Language	files	blank	comment	code
----	----	----	----	----
SUM:	38	980	557	8106

- ☑ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.
- There are six provided R files, four main and 11 function Matlab .m files, and 7 stata .do files
- Table 1, Table 3 and Figure 2 were made by the authors directly. There is no corresponding code file
- In-text numbers are not listed in the README, but a .do file is provided to replicate.

7.3 Computing Environment of the Replicator

- Nuvolos Cloud Server, "Ubuntu 22.04.3 LTS", Intel(R) Xeon(R) Platinum 8370C CPU @ 2.80GHz, 4 core, 16-GB RAM
- R version 4.5.1
- MATLAB 2025a
- STATA MP 19.5

```
R version 4.5.1 (2025-06-13)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 22.04.4 LTS

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK versi

locale:
 [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C               LC_TIME=en_US.UTF-8
 [4] LC_COLLATE=en_US.UTF-8     LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8       LC_NAME=C                  LC_ADDRESS=C
[10] LC_TELEPHONE=C             LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: Etc/UTC
tzcode source: system (glibc)

attached base packages:
[1] stats4    stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] kableExtra_1.4.0 dplyr_1.1.4      mvtnorm_1.3-3

loaded via a namespace (and not attached):
 [1] vctrs_0.6.5        svglite_2.2.1      cli_3.6.5          knitr_1.50         rlang_1.
 [6] xfun_0.52          stringi_1.8.7      generics_0.1.4     textshaping_1.0.1  glue_1.8
[11] htmltools_0.5.8.1  rsconnect_1.5.0    scales_1.4.0       rmarkdown_2.29     evaluate
[16] tibble_3.3.0       fastmap_1.2.0      lifecycle_1.0.4    stringr_1.5.1      compiler
```

```
[21] RColorBrewer_1.1-3 pkgconfig_2.0.3    rstudioapi_0.17.1  systemfonts_1.2.3  farver_2
[26] digest_0.6.37      viridisLite_0.4.2  R6_2.6.1           tidyselect_1.2.1   pillar_1
[31] magrittr_2.0.3     tools_4.5.1        withr_3.0.2        xml2_1.3.8
```

8 Replication

- Downloaded code and paper from Dropbox and GitHub to Nuvolos
- Changed STATA working directory byt replacing:

```
local folder "Desktop/replication_files" // update based on local directory (e.g. Desktop
    global user  "`c(username)'"
    if "`c(os)'" == "Windows" {
        global dir "C:/Users/$user/`folder'"
    }
    else if "`c(os)'" == "MacOSX" {
        global dir "/Users/$user/`folder'"
    }
```

with

```
global dir "/files/JPE-Sprenger-20250266/replication-package/replication_filesv2"
```

- Stata output ran with no other changes.
- Ran Matlab code as per instructions after changing directories. Worked without any further inputs.
- Ran R code as per instructions after changing directories. Worked without any further inputs.

9 Findings

- All outputs were reproduced. No code was provided for Table 1, Figure 2, and Table 3 since these were not computations.
- Tables D1, D2 and Figure D1 had different numbers for certain parameters (error was < 1.0, probably a computational issue)
- Data includes Personally Identifiable Information (PII) but PII has been anonymized to prevent identification. IRB approval included in the replication package.

9.1 Tables and Figures

Data Preparation Program	Dataset created	Reproduced?
clean_data.do	cleaned-data/ExperimentData.dta and .csv	
clean_data.do	cleaned-data/demographics.dta and .csv	
clean_data.do	cleaned-data/part1_cleaned.dta and .csv	

Data Preparation Program	Dataset created	Reproduced?
clean_data.do	cleaned-data/part2_cleaned.dta and .csv	
clean_data.do	cleaned-data/[dataset]_definitions.csv	

Figure/Table #	Program	Line Number	Reproduced?
Table 1	Made by Authors	N/A	N/A
Table 2	stata/makeTable2.do	91	yes
Table 3	Made by Authors	N/A	N/A
Figure 1	matlab/makeFigure1.m	133	yes
Figure 2	Made by Authors	N/A	N/A
Figure 3	stata/makeFigures3–7.do	16	yes
Figure 4	stata/makeFigures3–7.do	37	yes
Figure 5	stata/makeFigures3–7.do	64	yes
Figure 6	stata/makeFigures3–7.do	143	yes
Figure 7	stata/makeFigures3–7.do	222	yes
Figure 8	matlab/structuralEstimates.m	313	yes
Figure 9	matlab/makeFigure9.m	146	yes
Table A.1	stata/makeAppendixTables.do	21	yes
Table A.2	stata/makeAppendixTables.do	69	yes
Table A.3	stata/makeAppendixTables.do	92	yes
Table A.4	stata/makeAppendixTables.do	112	yes
Table A.5	stata/makeAppendixTables.do	154	yes
Table A.6	stata/makeAppendixTables.do	178	yes
Table A.7	stata/makeAppendixTables.do	207	yes
Table A.8	stata/makeAppendixTables.do	290	yes
Table A.9	stata/makeAppendixTables.do	436	yes

Figure/Table #	Program	Line Number	Reproduced?
Figure A.1	stata/makeAppendixFigures.do	17	yes
Figure A.2	stata/makeAppendixFigures.do	70	yes
Figure A.3	stata/makeAppendixFigures.do	119	yes
Figure A.4	stata/makeAppendixFigures.do	171	yes
Figure A.5	matlab/structuralEstimates.m	350	yes
Table C.1	stata/makeAppendixTables.do	509	yes
Figure C.1	stata/makeAppendixFigures.do	222	yes
Figure D.1	R/makeFigureD1.R	7	yes
Table D.1	R/makeTableD1.R	28	yes
Table D.2	R/makeTableD2.R	27	yes
Figure E.1	matlab/makeFigureE1.m	65	yes
Figure F.1	matlab/structuralEstimates.m	354	yes
Figure F.2	matlab/structuralEstimates.m	350	yes
Figure F.3	matlab/structuralEstimates.m	352	yes
Table F.1	matlab/structuralEstimates.m	426	yes
Figure G.1	matlab/structuralEstimates.m	813	yes
Figure G.2	matlab/structuralEstimates.m	813	yes
Figure G.3	matlab/structuralEstimates.m	813	yes
Figure G.4	matlab/structuralEstimates.m	813	yes
Figure G.5	matlab/structuralEstimates.m	813	yes
Figure G.6	matlab/structuralEstimates.m	813	yes
Table G.1	matlab/structuralEstimates.m	543	yes

In-text claim	Program	Line Number	Reproduced?
p. 11: 48 (34%) of 143 CR experiments use KT parameters	statsInText.do	20	Yes
p. 11: 34 (42%) of 81 CC experiments use Allais parameters	statsInText.do	20	Yes
p. 19: Average bonus payment $15.76	statsInText.do	20	Yes
p. 24: 21.0% of observations exhibit CR+CC	statsInText.do	37	Yes
p. 24: 14.6% exhibit modal pattern	statsInText.do	47	Yes
p. 24 (fn 31): 28.9% exhibit CR+CC near canonical parameters	statsInText.do	47	Yes
p. 24 (fn 31): 13.9% exhibit CR-RCC-MX near canonical parameters	statsInText.do	47	Yes
p. 24 (fn 31): 21.4% exhibit CR-RCC-MX at small p and small r	statsInText.do	47	Yes
p. 25 (fn 33): ~10% of observations at price list boundaries	statsInText.do	91	Yes
p. 26 (fn 38): 26% of response patterns are intransitive	statsInText.do	104	Yes
p. 26 (fn 38): 1.9% average per intransitive pattern	statsInText.do	104	Yes
p. 26 (fn 38): Only 2 intransitive patterns exceed 3%	statsInText.do	104	Yes
p. 27: 73% of simulated preferences in 4 strict patterns	statsInText.do	132	Yes
p. 27: 44% of raw responses in 4 strict patterns	statsInText.do	132	Yes
p. 27: 8% of simulated preferences in 3 weak patterns	statsInText.do	132	Yes
p. 27: 11% of raw responses in 3 weak patterns	statsInText.do	132	Yes
p. 27: 81% of simulated preferences in 7 marked patterns	statsInText.do	132	Yes
p. 27: 55% of raw responses in 7 marked patterns	statsInText.do	132	Yes
p. 32: 19% of simulated preferences in other 6 patterns	statsInText.do	175	Yes
p. 32: 18% of raw responses in other 6 transitive patterns	statsInText.do	175	Yes

In-text claim	Program	Line Number	Reproduced?
p. 33: ~13% of observations exhibit risk tolerance	statsInText.do	202	Yes
p. 33: 55% of risk-tolerant exhibit RCRP	statsInText.do	202	Yes
p. 33: 87% of observations exhibit risk aversion	statsInText.do	202	Yes
p. 33: 69% of risk-averse exhibit CRP	statsInText.do	202	Yes
p. 33: Fisher p < 0.001 (Risk Aversion x CRP)	statsInText.do	202	Yes
p. 33: 48% of risk-tolerant exhibit P1	statsInText.do	202	Yes
p. 33: 21% of risk-averse exhibit P1	statsInText.do	202	Yes
p. 33: Fisher p < 0.001 (Risk Aversion x P1)	statsInText.do	202	Yes
p. 33: 69% of risk-averse exhibit P2/P3/P4	statsInText.do	202	Yes
p. 33: 45% of risk-tolerant exhibit P2/P3/P4	statsInText.do	202	Yes
p. 33: Fisher p < 0.001 (Risk Aversion x P2/P3/P4)	statsInText.do	202	Yes

- Figure 1, 5 and 6: No labels, output same.
- Figure 2: no program provided
- Figure A1 - A3: same issue as Figure 6, image not provided for brevity
- Figure D1: slightly different from the paper (possibly computational differences). Comparison below:

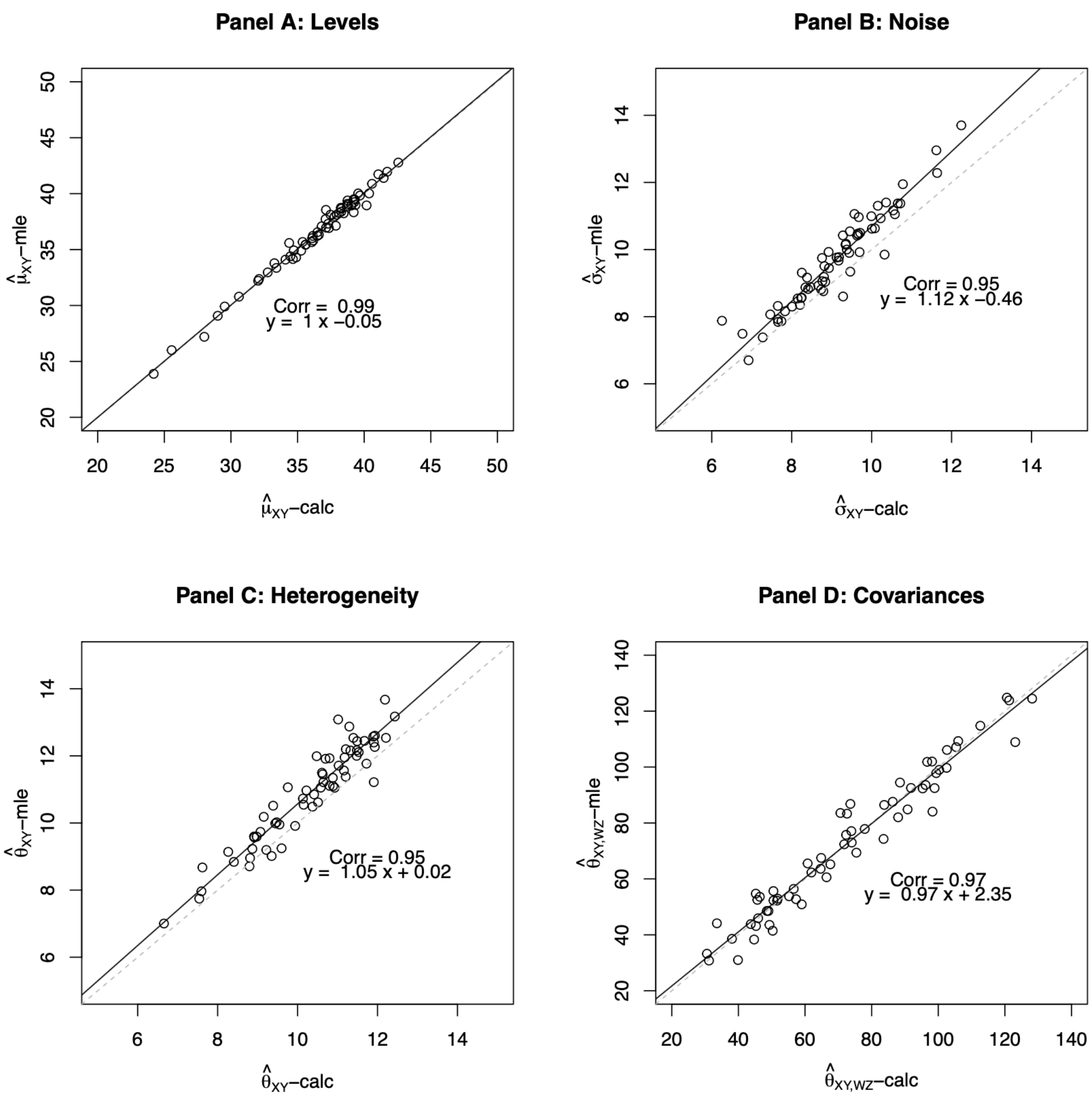
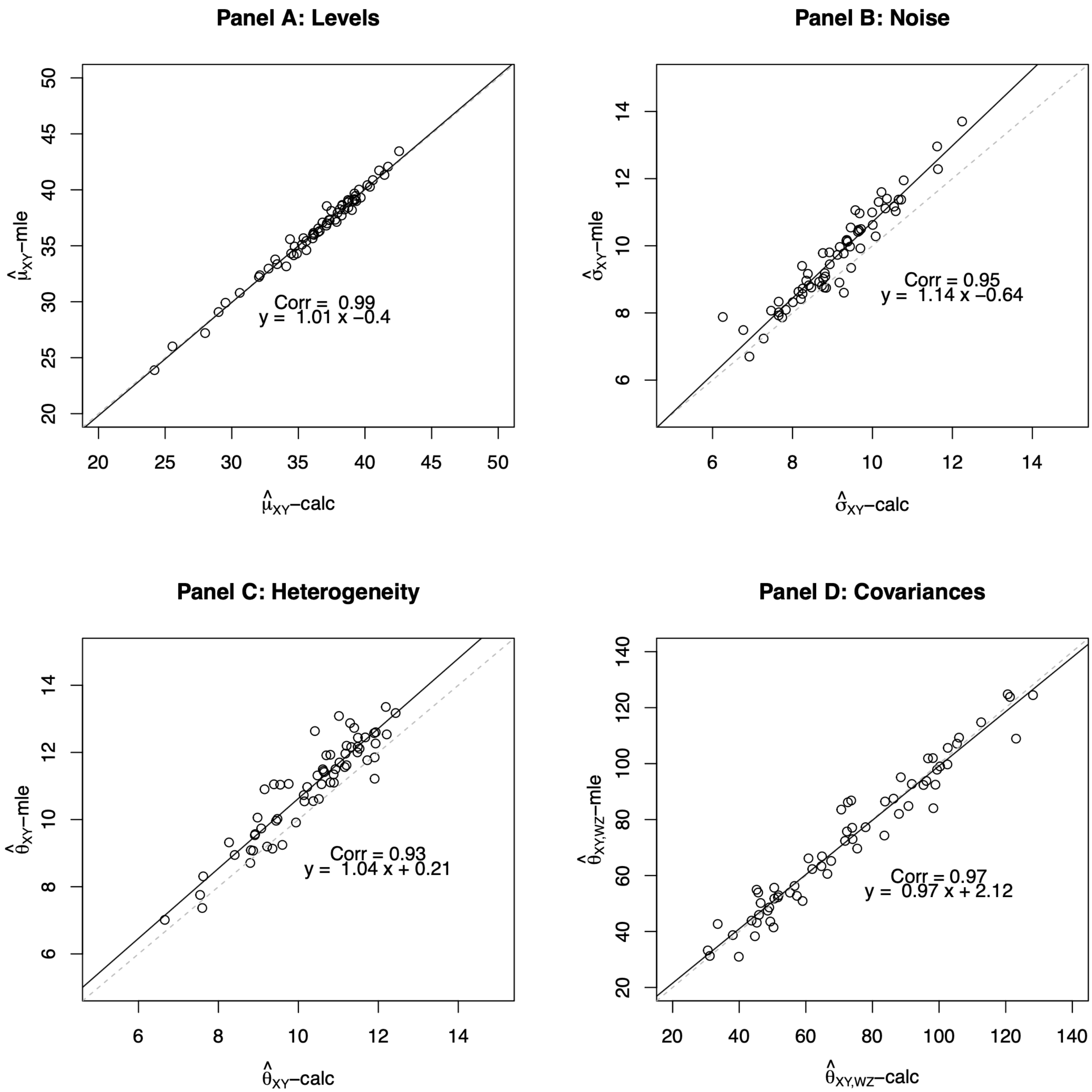


Figure D.1: Comparison of Decomposition Results (Direct Calculation vs. MLE)

Paper version



Replicated version

Figure 1

- Tables D1 and D2 have slight mismatch between numbers when is high. Outputs are marked below:

Table D.1: Decomposition Calculations (Levels)

p	r	$\hat{\mu}^*_{AB}$	$\hat{\mu}^*_{AB'}$	$\hat{\mu}^*_{CD}$	$\hat{\theta}^2_{AB}$	$\hat{\theta}^2_{AB'}$	$\hat{\theta}^2_{CD}$	$\hat{\sigma}^2_{AB}$	$\hat{\sigma}^2_{AB'}$	$\hat{\sigma}^2_{CD}$	$\hat{\theta}_{AB,AB'}$	$\hat{\theta}_{AB,CD}$	$\hat{\theta}_{AB',CD}$	$\frac{\widehat{var(h^*_{AB})}}{\widehat{var(h_{AB})}}$	$\frac{\widehat{var(h^*_{AB'})}}{\widehat{var(h_{AB'})}}$	$\frac{\widehat{var(h^*_{CD})}}{\widehat{var(h_{CD})}}$
0.30	0.10	36.32	23.89	32.96	165.72	142.30	171.21	87.27	129.38	187.73	65.22	75.75	50.90	0.66	0.52	0.4
0.30	0.20	35.68	26.01	32.37	147.88	148.81	132.19	111.17	109.70	150.86	86.45	74.29	52.92	0.57	0.58	0.4
0.30	0.30	36.55	29.08	35.59	154.91	173.54	158.78	110.22	108.48	130.01	114.77	99.71	84.06	0.58	0.62	0.5
0.30	0.50	38.12	30.79	38.27	131.12	100.42	123.34	98.47	167.92	120.87	107.12	101.86	83.57	0.57	0.37	0.5
0.30	0.80	38.55	34.93	37.08	123.18	125.81	112.63	79.66	89.28	120.25	108.92	97.83	92.47	0.61	0.58	0.4
0.50	0.10	37.13	27.20	32.21	144.02	128.73	111.15	56.11	102.89	129.35	60.59	30.98	52.74	0.72	0.56	0.4
0.50	0.20	39.09	29.91	33.36	141.93	84.57	120.34	76.70	142.77	112.80	73.01	52.22	43.15	0.65	0.37	0.5
0.50	0.30	38.99	33.78	35.43	138.49	146.69	75.83	62.08	77.64	124.71	84.85	41.46	43.56	0.69	0.65	0.3
0.50	0.50	38.97	34.15	40.02	158.14	157.17	122.18	61.86	73.42	84.33	124.47	109.29	101.97	0.72	0.68	0.5
0.50	0.80	38.65	37.31	38.24	133.73	150.45	98.27	44.89	84.00	77.84	123.83	92.35	99.00	0.75	0.64	0.5
0.80	0.10	41.73	35.66	34.30	122.38	148.05	91.53	73.99	122.34	127.81	82.03	33.24	55.65	0.62	0.55	0.4
0.80	0.20	39.46	36.02	36.23	85.48	154.69	94.78	65.08	62.81	103.52	77.09	45.94	72.38	0.57	0.71	0.4
0.80	0.30	41.96	39.24	35.81	142.87	157.14	78.15	81.71	122.15	95.29	93.58	54.73	53.78	0.64	0.56	0.4
0.80	0.50	39.01	36.96	37.95	80.20	109.95	59.97	72.98	69.23	98.05	65.54	38.58	38.32	0.52	0.61	0.3
0.80	0.80	40.88	41.40	42.78	129.48	153.71	81.27	54.50	73.24	90.39	124.83	77.86	69.38	0.70	0.68	0.4
0.90	0.10	40.02	35.44	34.10	117.75	103.71	83.53	108.62	119.50	98.60	83.33	30.79	56.43	0.52	0.46	0.4
0.90	0.20	39.83	38.93	34.90	122.09	143.76	75.25	86.71	97.02	95.03	106.11	44.12	48.54	0.58	0.60	0.4
0.90	0.30	38.33	38.95	36.92	92.02	187.01	92.09	82.05	68.90	95.55	94.46	52.51	67.51	0.53	0.73	0.4
0.90	0.50	39.39	38.74	37.72	99.10	110.57	63.43	93.54	113.06	78.71	86.81	43.83	53.60	0.51	0.49	0.4
0.90	0.80	38.39	39.50	38.11	85.16	115.05	49.05	69.71	66.60	61.51	87.58	48.53	52.27	0.55	0.63	0.4
0.62	0.38	38.85	34.39	36.22	125.78	137.11	99.75	78.87	100.02	109.16	92.53	62.29	63.61	0.61	0.58	0.4

Note: Decomposition estimates from MLE exercise. Final line presents averages over all 20 rows.

47

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Table 1: Decomposition Calculations (Levels)

p	r	$\hat{\mu}^*_{AB}$	$\hat{\mu}^*_{AB'}$	$\hat{\mu}^*_{CD}$	$\hat{\theta}^2_{AB}$	$\hat{\theta}^2_{AB'}$	$\hat{\theta}^2_{CD}$	$\hat{\sigma}^2_{AB}$	$\hat{\sigma}^2_{AB'}$	$\hat{\sigma}^2_{CD}$	$\hat{\theta}_{AB,AB'}$	$\hat{\theta}_{AB,CD}$	$\hat{\theta}_{AB',CD}$	$\frac{\widehat{var(h^*_{AB})}}{\widehat{var(h_{AB})}}$	$\frac{\widehat{var(h^*_{AB'})}}{\widehat{var(h_{AB'})}}$	$\frac{\widehat{var(h^*_{CD})}}{\widehat{var(h_{CD})}}$
0.30	0.10	36.32	23.89	32.96	165.72	142.30	171.21	87.27	129.38	187.73	65.22	75.75	50.90	0.66	0.52	0.48
0.30	0.20	35.68	26.01	32.37	147.88	148.81	132.19	111.17	109.70	150.86	86.45	74.29	52.92	0.57	0.58	0.47
0.30	0.30	36.55	29.08	35.59	154.91	173.54	158.78	110.22	108.48	130.01	114.77	99.71	84.06	0.58	0.62	0.55
0.30	0.50	38.12	30.79	38.27	131.12	100.42	123.34	98.47	167.92	120.87	107.12	101.86	83.57	0.57	0.37	0.51
0.30	0.80	38.55	34.93	37.08	123.18	125.81	112.63	79.66	89.28	120.25	108.92	97.83	92.47	0.61	0.58	0.48
0.50	0.10	37.13	27.20	32.21	144.02	128.73	111.15	56.11	102.89	129.35	60.59	30.98	52.74	0.72	0.56	0.46
0.50	0.20	39.09	29.91	33.36	141.93	84.57	120.34	76.70	142.77	112.80	73.01	52.22	43.15	0.65	0.37	0.52
0.50	0.30	38.99	33.78	35.43	138.49	146.69	75.83	62.08	77.64	124.71	84.85	41.46	43.56	0.69	0.65	0.38
0.50	0.50	38.97	34.15	40.02	158.14	157.17	122.18	61.86	73.42	84.33	124.47	109.29	101.97	0.72	0.68	0.59
0.50	0.80	38.65	37.31	38.24	133.73	150.45	98.27	44.89	84.00	77.84	123.83	92.35	99.00	0.75	0.64	0.56
0.80	0.10	41.73	35.66	34.30	122.38	148.05	91.53	73.99	122.34	127.81	82.03	33.24	55.65	0.62	0.55	0.42
0.80	0.20	39.46	36.02	36.23	85.48	154.69	94.78	65.08	62.81	103.52	77.09	45.94	72.38	0.57	0.71	0.48
0.80	0.30	42.07	39.09	35.97	143.13	162.05	80.02	76.51	121.64	94.65	93.80	54.93	53.89	0.65	0.57	0.46
0.80	0.50	38.99	37.00	37.33	82.59	111.30	60.15	74.53	69.49	99.53	66.14	38.71	38.31	0.53	0.62	0.38
0.80	0.80	40.87	41.33	43.45	135.00	140.46	83.42	52.38	88.37	82.50	124.78	77.27	69.64	0.72	0.61	0.50
0.90	0.10	40.27	34.61	33.16	159.67	118.85	86.85	95.48	134.54	96.10	86.17	31.27	56.28	0.63	0.47	0.47
0.90	0.20	39.30	38.19	35.10	132.29	128.01	69.05	76.14	123.48	95.71	105.62	42.68	47.46	0.63	0.51	0.42
0.90	0.30	39.67	40.42	37.28	101.17	178.33	90.89	81.31	69.20	99.32	95.12	53.92	66.86	0.55	0.72	0.48
0.90	0.50	38.41	37.71	36.80	121.85	122.10	54.25	79.28	105.70	80.33	86.82	43.89	50.18	0.61	0.54	0.40
0.90	0.80	38.58	39.17	37.94	82.33	115.09	49.18	70.79	65.52	64.31	87.48	48.59	51.75	0.54	0.64	0.43
0.62	0.38	38.87	34.31	36.15	130.25	136.87	99.30	76.70	102.43	109.13	92.71	62.31	63.34	0.63	0.58	0.47

Note: Decomposition estimates from MLE exercise. Final line presents averages over all 20 rows.

Replicated version

Figure 2

Table D.2: Decomposition Calculations (Differences)

p	r	$\hat{\Delta}^{**}_{CR}$	$\hat{\Delta}^{**}_{CC}$	$\hat{\Delta}^{**}_{MX}$	$\widehat{var(\Delta^*_{CR})}$	$\widehat{var(\Delta^*_{CC})}$	$\widehat{var(\Delta^*_{MX})}$	$\widehat{var(\Delta_{CR})}$	$\widehat{var(\Delta_{CC})}$	$\widehat{var(\Delta_{MX})}$	$\frac{\widehat{var(\Delta^*_{CR})}}{\widehat{var(\Delta_{CR})}}$	$\frac{\widehat{var(\Delta^*_{CC})}}{\widehat{var(\Delta_{CC})}}$	$\frac{\widehat{var(\Delta^*_{MX})}}{\widehat{var(\Delta_{MX})}}$
0.30	0.10	3.36	-9.07	12.43	185.44	211.71	177.59	460.44	528.82	394.23	0.40	0.40	0.45
0.30	0.20	3.31	-6.36	9.67	131.49	175.15	123.79	393.52	435.72	344.67	0.33	0.40	0.36
0.30	0.30	0.96	-6.51	7.47	114.26	164.18	98.90	354.49	402.68	317.60	0.32	0.41	0.31
0.30	0.50	-0.14	-7.48	7.34	50.74	56.61	17.30	270.08	345.40	283.69	0.19	0.16	0.06
0.30	0.80	1.47	-2.16	3.62	40.16	53.50	31.15	240.06	263.03	200.09	0.17	0.20	0.16
0.50	0.10	4.93	-5.01	9.93	193.21	134.40	151.56	378.67	366.64	310.57	0.51	0.37	0.49
0.50	0.20	5.73	-3.45	9.18	157.85	118.61	80.49	347.34	374.18	299.96	0.45	0.32	0.27
0.50	0.30	3.56	-1.65	5.21	131.40	135.41	115.48	318.19	337.75	255.20	0.41	0.40	0.45
0.50	0.50	-1.05	-5.87	4.82	61.74	75.41	66.37	207.93	233.17	201.65	0.30	0.32	0.33
0.50	0.80	0.41	-0.93	1.34	47.30	50.71	36.51	170.03	212.55	165.40	0.28	0.24	0.22
0.80	0.10	7.44	1.36	6.07	147.44	128.28	106.38	349.24	378.43	302.70	0.42	0.34	0.35
0.80	0.20	3.23	-0.21	3.44	88.38	104.71	86.00	256.97	271.03	213.89	0.34	0.39	0.40
0.80	0.30	6.15	3.43	2.72	111.56	127.73	112.85	288.56	345.16	316.71	0.39	0.37	0.36
0.80	0.50	1.06	-0.99	2.05	63.01	93.28	59.06	234.05	260.56	201.27	0.27	0.36	0.29
0.80	0.80	-1.90	-1.38	-0.52	55.03	96.23	33.52	199.91	259.85	161.26	0.28	0.37	0.21
0.90	0.10	5.92	1.34	4.58	139.70	74.38	54.81	346.92	292.48	282.93	0.40	0.25	0.19
0.90	0.20	4.93	4.03	0.90	109.09	121.93	53.62	290.83	313.98	237.35	0.38	0.39	0.23
0.90	0.30	1.41	2.02	-0.62	79.09	144.08	90.12	256.69	308.54	241.07	0.31	0.47	0.37
0.90	0.50	1.68	1.02	0.66	74.86	66.80	36.06	247.12	258.57	242.67	0.30	0.26	0.15
0.90	0.80	0.28	1.39	-1.10	37.15	59.56	25.06	168.37	187.67	161.37	0.22	0.32	0.16
0.62	0.38	2.64	-1.82	4.46	100.94	109.63	77.83	288.97	318.81	256.71	0.33	0.34	0.29

Note: Decomposition estimates calculated from from MLE exercise. Final line presents averages over all 20 rows.

48

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Table 2: Decomposition Calculations (Differences)

p	r	$\hat{\Delta}^{**}_{CR}$	$\hat{\Delta}^{**}_{CC}$	$\hat{\Delta}^{**}_{MX}$	$\widehat{var(\Delta^*_{CR})}$	$\widehat{var(\Delta^*_{CC})}$	$\widehat{var(\Delta^*_{MX})}$	$\widehat{var(\Delta_{CR})}$	$\widehat{var(\Delta_{CC})}$	$\widehat{var(\Delta_{MX})}$	$\frac{\widehat{var(\Delta^*_{CR})}}{\widehat{var(\Delta_{CR})}}$	$\frac{\widehat{var(\Delta^*_{CC})}}{\widehat{var(\Delta_{CC})}}$	$\frac{\widehat{var(\Delta^*_{MX})}}{\widehat{var(\Delta_{MX})}}$
0.30	0.10	3.36	-9.07	12.43	185.44	211.71	177.59	460.44	528.82	394.23	0.40	0.40	0.45
0.30	0.20	3.31	-6.36	9.67	131.49	175.15	123.79	393.52	435.72	344.67	0.33	0.40	0.36
0.30	0.30	0.96	-6.51	7.47	114.26	164.18	98.90	354.49	402.68	317.60	0.32	0.41	0.31
0.30	0.50	-0.14	-7.48	7.34	50.74	56.61	17.30	270.08	345.40	283.69	0.19	0.16	0.06
0.30	0.80	1.47	-2.16	3.62	40.16	53.50	31.15	240.06	263.03	200.09	0.17	0.20	0.16
0.50	0.10	4.93	-5.01	9.93	193.21	134.40	151.56	378.67	366.64	310.57	0.51	0.37	0.49
0.50	0.20	5.73	-3.45	9.18	157.85	118.61	80.49	347.34	374.18	299.96	0.45	0.32	0.27
0.50	0.30	3.56	-1.65	5.21	131.40	135.41	115.48	318.19	337.75	255.20	0.41	0.40	0.45
0.50	0.50	-1.05	-5.87	4.82	61.74	75.41	66.37	207.93	233.17	201.65	0.30	0.32	0.33
0.50	0.80	0.41	-0.93	1.34	47.30	50.71	36.51	170.03	212.55	165.40	0.28	0.24	0.22
0.80	0.10	7.44	1.36	6.07	147.44	128.28	106.38	349.24	378.43	302.70	0.42	0.34	0.35
0.80	0.20	3.23	-0.21	3.44	88.38	104.71	86.00	256.97	271.03	213.89	0.34	0.39	0.40
0.80	0.30	6.10	3.12	2.98	113.29	134.29	117.58	284.45	350.59	315.73	0.40	0.38	0.37
0.80	0.50	1.66	-0.34	1.99	65.32	94.82	61.60	239.37	263.84	205.61	0.27	0.36	0.30
0.80	0.80	-2.58	-2.12	-0.46	63.88	84.59	25.89	198.75	255.47	166.64	0.32	0.33	0.16
0.90	0.10	7.11	1.45	5.66	183.99	93.15	106.19	375.57	323.79	336.20	0.49	0.29	0.32
0.90	0.20	4.20	3.09	1.12	115.98	102.14	49.07	287.82	321.32	248.69	0.40	0.32	0.20
0.90	0.30	2.39	3.14	-0.75	84.21	135.51	89.25	264.84	304.02	239.76	0.32	0.45	0.37
0.90	0.50	1.61	0.92	0.69	88.31	75.99	70.32	247.92	262.02	255.30	0.36	0.29	0.28
0.90	0.80	0.64	1.23	-0.59	34.34	60.78	22.45	169.43	190.61	158.76	0.20	0.32	0.14
0.62	0.38	2.72	-1.84	4.56	104.94	109.50	81.69	290.76	321.05	260.82	0.34	0.33	0.30

Note: Decomposition estimates calculated from from MLE exercise. Final line presents averages over all 20 rows.

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Figure 3

- Other figures and tables match the reported output in the paper

9.2 In-Text Numbers

- [x] In-text numbers verified.

10 Classification

- [ ] full reproduction
- [x] full reproduction with minor issues
- [ ] partial reproduction (see above)
- [ ] not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- [x] None.
- [ ] `Discrepancy in output` (either figures or numbers in tables or text differ)
- [ ] `Bugs in code` that were fixable by the replicator (but should be fixed in the final deposit)

- [ ] `Code missing`, in particular if it prevented the replicator from completing the reproducibility check
    - [ ] `Data preparation code missing` should be checked if the code missing seems to be data preparation code
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- [ ] `Data missing` is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
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